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**MID TERM EXAMINATION**

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)  
3rd Semester, November, 2022

**Paper Code: ECC-207****Subject: Digital Logic and Computer Design****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.**

| Q. No.        | Question                                                                                                                                                                                                   | Max. Marks | CO(s)         |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------|
| <b>1. (a)</b> | How is BCD Code different from Binary Code?                                                                                                                                                                | <b>2</b>   | <b>CO1</b>    |
| <b>(b)</b>    | Explain addition of two negative numbers in 2's complement form with the help of an example.                                                                                                               | <b>2</b>   | <b>CO1</b>    |
| <b>(c)</b>    | Implement a two-input EX-OR gate using a Multiplexer.                                                                                                                                                      | <b>2</b>   | <b>CO1</b>    |
| <b>(d)</b>    | Explain the concept of Level Triggering in Sequential Circuits.                                                                                                                                            | <b>2</b>   | <b>CO2</b>    |
| <b>(e)</b>    | How is a state diagram different from a state table?                                                                                                                                                       | <b>2</b>   | <b>CO2</b>    |
| <b>2. (a)</b> | Differentiate between Serial Adder and Binary Parallel Adder.                                                                                                                                              | <b>3</b>   | <b>CO1, 2</b> |
| <b>(b)</b>    | Explain how to convert an SR Flip-Flop into a D Flip-Flop.                                                                                                                                                 | <b>3</b>   | <b>CO2</b>    |
| <b>(c)</b>    | Design a 4-bit Ripple Down Counter and draw the waveform.                                                                                                                                                  | <b>4</b>   | <b>CO2</b>    |
| <b>3. (a)</b> | For the Boolean expression $F = A'B'C' + A'BC' + AB'C' + ABC + ABC'$ , find out the minimized Product of Sum (POS) expression using K-map.<br>[Note: Prime notation denotes logical negation / inversion]. | <b>5</b>   | <b>CO1</b>    |
| <b>(b)</b>    | What is a Multiplexer? Design a 16:1 multiplexer using 4:1 multiplexers.                                                                                                                                   | <b>5</b>   | <b>CO1</b>    |
| <b>4. (a)</b> | Using K-map, simplify the expression: $Y = f(A, B, C, D) = \prod M(0, 1, 2, 4, 5, 8, 9, 15)$ .                                                                                                             | <b>2</b>   | <b>CO1</b>    |
| <b>(b)</b>    | A combinational circuit is defined by functions $F_1 = \sum m(1, 5, 7)$ and $F_2 = \sum m(5, 6, 7)$ . Implement the circuit with a PLA.                                                                    | <b>3</b>   | <b>CO2</b>    |
| <b>(c)</b>    | Design a Mod-5 asynchronous counter using T Flip-Flops.                                                                                                                                                    | <b>5</b>   | <b>CO2</b>    |

**\*\*\*\* End of Question Paper \*\*\*\***